

Samarvir Garg

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EDUCATION

University of Toronto

Bachelor of Applied Science and Engineering - Computer Engineering

Toronto, ON

Sep. 2025 – Present

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS

Frameworks: Spring Boot, TensorFlow, REST APIs

Developer Tools: Git, AWS, Power BI, VS Code, IntelliJ, Arduino

Libraries: OpenAI API, NumPy, Pandas

EXPERIENCE

AI Engineer Trainee

May. 2024 – Present

Trakop - Retail Supply and logistics platform

Toronto, ON

- Developing AI based solutions for logistics and business automation using Python, TensorFlow, and OpenAI APIs
- Working on projects related to inventory forecasting, delivery route optimization, and payment risk analysis
- Gaining hands on experience with AWS, Power BI, and Git while contributing to real world SaaS platform development

Member, Driverless – Deep Perception Subteam

Jan 2026 – Present

University of Toronto Formula Racing

Toronto, ON

- Working on ML-based perception for cone detection, classification, and localization using camera image streams.
- Contributing to data collection and manual labeling to improve model generalization.
- Evaluating models using IoU, mAP, and inference latency under real-time NVIDIA Jetson constraints.

Research Assistant

Nov. 2024 – Feb. 2025

University of Toronto (as a High School Research Intern)

Toronto, ON

- Conducted research under Prof. Clement C. Zai on "Pharmacogenetic Reporting and Data Visualization in Mental Healthcare"
- Developed AI-powered clinical dashboards and decision-support tools using LLMs to automate pharmacogenetic data reporting
- Authored a technical report proposing scalable, data-driven visualization design recommendations.

PROJECTS

Emergency Route Optimization System | Java, Spring Boot, Becker Robots, SQLite, REST API

Jan. 2026

- Designed and developed a full-stack backend system using Java Spring Boot following a 3-layer (Controller–Service–Repository) architecture
- Developed a REST API with CRUD operations to handle emergency inputs, ambulance updates, and routing decisions
- Integrated a SQLite database for persistence of emergency data and simulation logs, and validated logic through a Java-based simulation frontend using Becker Robots; placed 3rd overall at UTEK 2026

Autonomous Robotics System | C++, Arduino Uno R4 Minima

Jan. 2026

- Built an autonomous Arduino robot for a competition course involving navigation, object pickup, and target alignment
- Integrated IR sensors, ultrasonic ranging, and color sensing for line following, obstacle detection, and path-based behavior selection
- Implemented a modular C++ software architecture separating hardware control, sensor perception, and high-level behaviors to improve reliability and debugging speed
- Placed 2nd Overall and 1st for Top Track Performance out of 60+ teams at UTRA HACKS 2026